

Grayscale Gradation Representation for Direction-Dependent Visualization using High-Speed Projection

Hiroto Tsutsumoto*
Institute of Science Tokyo

Ryusuke Miyazaki†
Institute of Science Tokyo

Shio Miyafuji‡
Institute of Science Tokyo

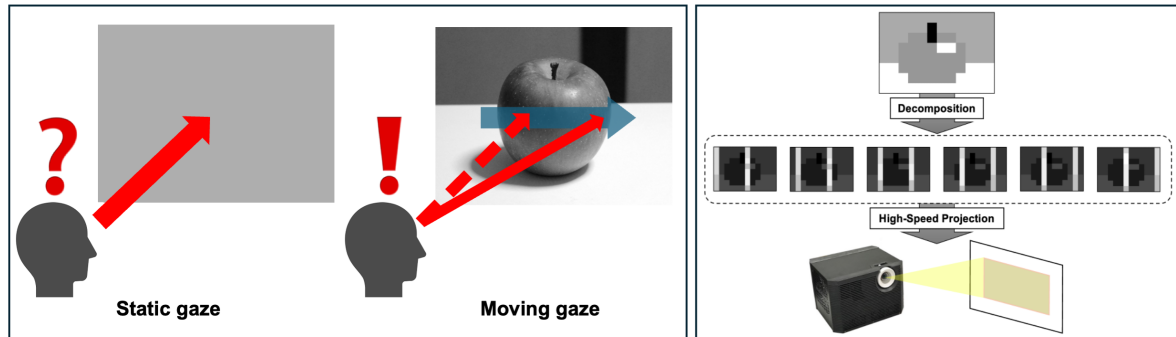


Figure 1: (Left) The grayscale image is perceived only when the gaze moves in a specific direction. (Right) Proposed method. The image is decomposed into stripe patterns and projected at a high frame rate.

ABSTRACT

Projection-based information display offers the advantage of presenting information to multiple users simultaneously using a single device. However, because all users inevitably view the same projected content, it is difficult to provide personalized information tailored to individual users. To overcome this limitation, prior work has introduced high-speed projection techniques that display different images depending on the observer's gaze movement direction. These methods require neither tracking devices nor user-specific hardware; instead, the perceived content changes solely through the observer's own gaze motion. Nevertheless, the presentable images have been restricted to binary images. In this study, we extend this paradigm by enabling grayscale representation, and we propose a method for achieving richer visual presentation for direction-dependent visualization.

Index Terms: Spatial augmented reality, High-speed projection, Gaze movement, Temporal additive color mixing

1 INTRODUCTION

Projection-based information display has the advantage of allowing a single device to simultaneously present information to multiple users, making it suitable for use in public spaces. However, because all users observe the same content, it is difficult to personalize the presented information for individual users.

To achieve personalized presentation using projectors, interaction techniques have been proposed, such as hand-gesture tracking [2] and approaches that rely on auxiliary devices [8][9]. However, approaches that depend on tracking technologies or additional devices are not suitable for public environments in which an unspecified number of viewers may be present.

*e-mail: tsutsumoto.h.bd78@m.isct.ac.jp

†e-mail: miyazaki.r.a5eb@m.isct.ac.jp

‡e-mail: miyafuji@comp.isct.ac.jp

As a tracking-free approach, Sakaue et al. [7] proposed a method that changes the projected image depending on the distance between the projector and the screen. Although this method realizes view-dependent image variation, it is limited to movement along the screen normal direction. Ikeda et al. [4] proposed a method that shifts the projected image laterally to generate different perceived images, but the projected pattern does not convey any semantic information when the viewer remains stationary.

To address this issue, techniques exploiting high-speed projection have recently been proposed [5][6]. These methods decompose an image into stripe patterns and display them at high speed, enabling the encoded image to become visible only during eye movements, while remaining imperceptible when the gaze is static. However, images supported by prior work were limited to binary content composed of black-and-white patterns.

We propose a method for grayscale representation that compresses the image intensity range and superimposes stripes with complementary luminance. This approach increases the expressive capability of direction-dependent display using high-speed projection. Furthermore, we conducted a user study to evaluate the perceptual characteristics of the proposed grayscale representation and investigated how the number of frame divisions affects the perceptible luminance differences. Finally, we describe the application of an interactive display system that switches the displayed content according to the observer's gaze state.

2 METHOD

2.1 Overview of the Proposed Method

In prior work, white stripe patterns were added to black regions of the original image so that a uniform gray field could be perceived when the viewer's gaze was static. The stripe patterns were rendered at high speed such that they became perceptible only during eye movements, allowing users to recognize the intended image structure. However, the images handled in prior studies were limited to binary black-and-white patterns, and the display of general images containing grayscale tones was not considered.

In this work, we extend the conventional approach to support grayscale images containing luminance variations. Specifically, we

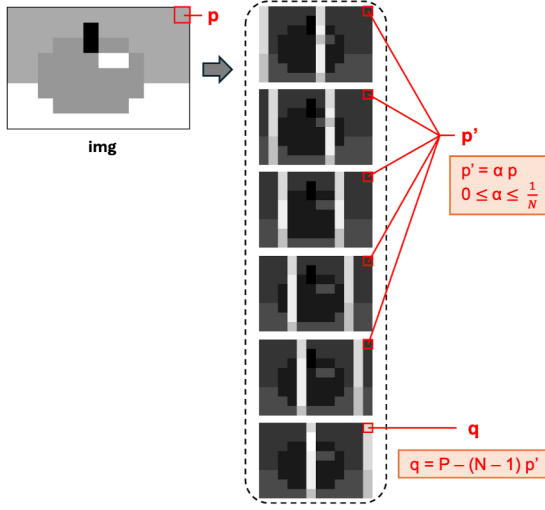


Figure 2: Image decomposition method

hypothesize that grayscale information can be presented during eye movements by superimposing stripe patterns of complementary luminance to each pixel value, while preserving a uniform appearance during static gaze. This enables a much wider expressive range compared to previous techniques that were restricted to binary images. To achieve this, we compress the pixel-value range of the original image toward the lower-luminance side, thereby restricting the luminance range of the original content to darker values, while assigning higher-luminance values to the added stripe patterns. By separating the luminance ranges of the original image and the stripe patterns, the stripe patterns become more distinct, leading to improved perceptual visibility of the original image during eye movements.

2.2 Image Decomposition Method

This subsection describes the image decomposition method used in the proposed technique. Let the original image be *img*, and let each pixel value take an integer within the range 0 to P . In this paper, we refer to this range of pixel values as the *dynamic range* of the image. The proposed method first compresses the dynamic range of the original image toward the low-luminance side. This allows the pixel values of the added complementary luminance stripe component to be allocated to the high-luminance side, thereby separating the luminance bands of the original image and the stripe component. Let p be an arbitrary pixel value in the original image, and let p' be the compressed pixel value. We define the dynamic range compression ratio α as a scaling factor satisfying $0 < \alpha \leq 1$. Then, p' is given by

$$p' = \alpha p \quad (1)$$

Next, the image is decomposed into N frames using an integer N . For each pixel, the value p' is assigned to $N - 1$ of the N frames, and the opposite-color component q is assigned to the remaining frame to form the stripe pattern. The complementary luminance component q is defined as

$$q = P - (N - 1)p' \quad (2)$$

The perceived pixel value during static gaze can be expressed as the mean value

$$\frac{(N - 1)p' + q}{N} \quad (3)$$

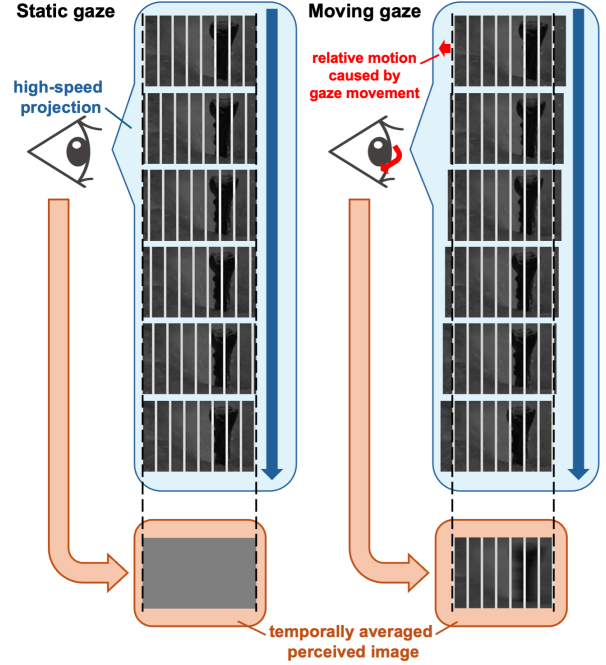


Figure 3: Principle of Direction-Dependent Visualization

Substituting (2) into (3) yields

$$\frac{(N - 1)p' + q}{N} = \frac{P}{N} \quad (4)$$

indicating that the pixel value becomes constant.

Furthermore, to separate the luminance bands of the original image and the stripe component, it is necessary to ensure $p' < q$. The maximum value of p' is αP , and the minimum value of q is $P - (N - 1)\alpha P$, leading to

$$\alpha P < P - (N - 1)\alpha P \quad (5)$$

Rearranging this inequality gives

$$\alpha \leq \frac{1}{N} \quad (6)$$

An overview of this decomposition method is shown in Figure 2.

2.3 High-speed Projection and Direction-dependent Visualization

In this section, we describe the projection process of the decomposed frames and the principle of direction-dependent visualization. An overview of the principle is shown in Figure 3.

The human visual system has a temporal frequency threshold for perceivable flicker, which is generally reported to be around 60 Hz[3]. When the display refresh rate exceeds this threshold, individual frames are no longer perceived separately.

In the proposed system, the decomposed image frames are sequentially projected at a high refresh rate such that the luminance of each pixel is temporally integrated in perception. The stripe patterns are designed such that the temporal average luminance of each pixel becomes constant; therefore, when the observer keeps the gaze stationary, the projected image is perceived as a uniform single-color image.



Figure 4: Projection characteristics of DynaFlash

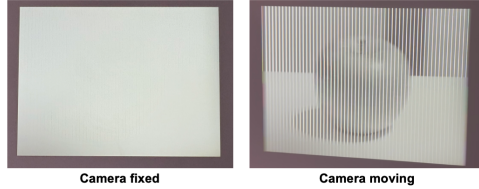


Figure 5: Projected image captured by a camera

In contrast, when the observer moves the gaze horizontally, the projected screen is perceived to move in the opposite direction in the observer's visual coordinate system. If the relative motion speed matches the motion speed of the stripe patterns, the stripe components of successive frames overlap spatially in the visual field. As a result, the observer can perceive the stripe patterns during gaze movement, allowing the original shape and grayscale information of the target image to be recognized.

3 IMPLEMENTATION

Since the input image used in our method is an 8-bit grayscale image, we set $P = 255$. Regarding the number of decomposition frames N , existing studies have used $N = 4$ for binary patterns. However, our method deals with images containing grayscale variations, requiring wider spacing between complementary luminance stripe components to ensure that the original image can be perceived during gaze motion. Therefore, in addition to $N = 4$, we implemented $N = 6$ and $N = 8$, and compared them through a user study. For the dynamic range compression factor α , we adopted a common parameter of $\alpha = 1/8$ for all values of N , so as to satisfy the constraint expressed in Equation (6). Image generation and preprocessing were performed using OpenCV.

The generated stripe patterns were repeatedly projected using a high-speed projector (Dynaflash¹) at 60 Hz. Due to the projection characteristics of Dynaflash, the projected images appear significantly brighter than the original digital images. Therefore, even with the pixel values reduced to $1/8$, the original content remained easily perceptible (Figure 4). Figure 5 shows the projection results of the proposed method, captured with a stationary camera and with a moving camera simulating eye motion. When captured with a stationary camera, the projected image appeared almost uniform and effectively concealing the underlying content. In contrast, when the camera was moved, coarse luminance variations became distinguishable, allowing observers to identify the displayed apple image.

4 EXPERIMENT

To evaluate the perception of tonal variations in direction-dependent display, we conducted a preliminary study with 12 participants (8 male, 4 female, aged 21–34). The experiment was performed under three stripe-division conditions ($N = 4, 6, 8$). As N

¹<https://www.inrevium.com/product/tb-6v-dynaflash/>, accessed on January 15, 2026

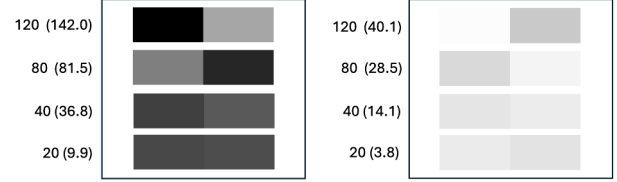


Figure 6: Images presented during eye movements. The left-right positions were randomized in each trial. The value shown to the left of each image indicates the pixel value difference, and the value in parentheses denotes the corresponding difference in projected luminance.

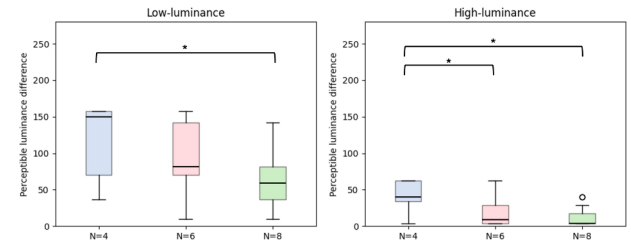


Figure 7: Boxplots of perceptible contrast differences measured in the experiment.

increases, the projection frequency becomes higher, requiring faster eye movements. On the other hand, a larger N provides larger spatial interval between stripes, potentially improving the perceptual discrimination of tonal differences in the target image.

The aim of this experiment was to measure the perceptible luminance difference. Participants were asked to track a horizontally moving pointer with their eyes, which was implemented such that images similar to those shown in Figure 6 became visible when following the pointer. The pointer's speed and direction matched the movement of the stripes. The presented luminance differences were defined as pixel value differences of $\{120, 80, 40, 20\}$, and both low-luminance and high-luminance conditions were evaluated. Each participant performed the task under three display conditions ($N = 4, 6, 8$). Participants were asked to indicate which side appeared darker between the left and right stimuli, with three response options: "left is darker," "right is darker," and "cannot determine." For each participant, the smallest luminance difference that was correctly identified was defined as the perceptible luminance difference. If a participant failed to answer correctly from the first trial, a pixel value difference of 160 was assigned based on the step size. This value was confirmed in a pilot study to be within a range that is subjectively but reliably perceptible.

To account for the luminance characteristics of the projector, the projected luminance values measured in advance using a camera were used as the ground truth values in this experiment. The results are shown in Figure 7. In the low-luminance condition, the Friedman test revealed a significant main effect of the division number N ($\chi^2 = 9.71, p = 0.0078$). Post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Holm correction showed a significant difference only between $N = 4$ and $N = 8$ ($p = 0.0176$), indicating that increasing N tends to enable the perception of images with smaller luminance differences. Similarly, in the high-luminance condition, a significant main effect of N was observed ($\chi^2 = 10.22, p = 0.0060$). Post-hoc tests revealed significant differences between $N = 4$ and $N = 6$ ($p = 0.0156$), and between $N = 4$ and $N = 8$ ($p = 0.0117$), suggesting that increasing N improves the perceptibility of smaller luminance differences.

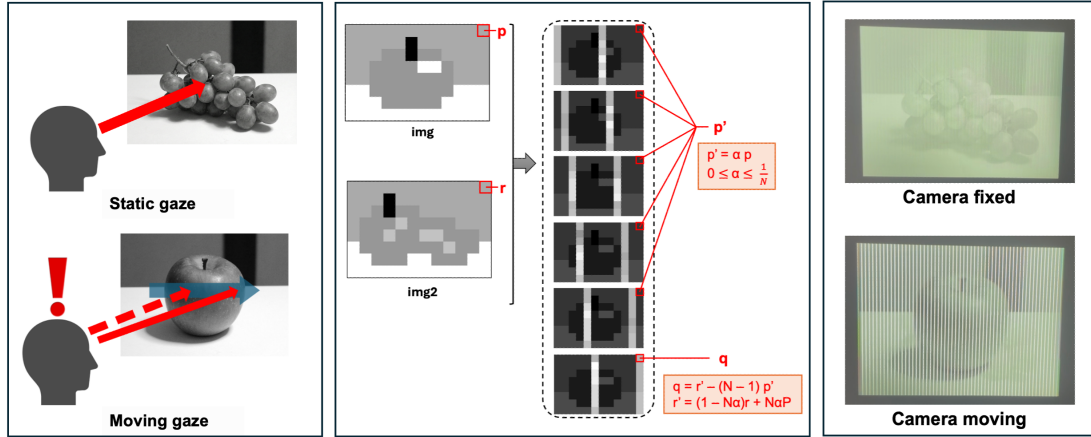


Figure 8: A method that presents two different images during eye fixation and during eye movements: (Left) Proposed method. (Center) Image decomposition method. (Right) Projected image captured by a camera.

5 DISCUSSION

The experimental results showed that the number of divisions N had a significant effect on the perceptually perceptible luminance differences for both the low-luminance and high-luminance conditions. This suggests that increasing the number of divisions allows wider spacing between stripe patterns of the opponent color components, making it easier to discriminate luminance differences in the target image. Furthermore, at least up to $N = 8$, the effect of increasing the proportion of the target image that becomes perceptible outweighs the negative effect of the increased eye movement speed required by the higher projection frequency.

In addition, the medians of the perceptible luminance differences at $N = 8$ were 59.1 for the low-luminance condition and 3.8 for the high-luminance condition. The high-luminance condition consistently exhibited smaller minimum perceptually perceptible luminance differences. It is known that the human visual system shows higher contrast sensitivity under higher mean luminance conditions, and this characteristic may have been reflected in our results [1]. Moreover, since the background image was fixed to white, relative perceptual differences caused by the background may also have influenced the results; therefore, additional experiments with different background images are required.

In this experiment, the differences in projected luminance obtained from the preliminary measurements were used as the ground-truth criteria for correct answers; however, these values are nonlinear and non-uniform due to the luminance characteristics of the projector. Therefore, validation under a wider range of finer luminance difference conditions is required.

This method has several limitations. First, in regions where the luminance differences are smaller than the perceptual threshold, it becomes difficult to discriminate the presented images. In particular, in low-luminance regions, fine luminance differences were found to be difficult to perceive. Second, to facilitate the perception of the presented images, we displayed a pointer on the screen to guide the observer's eye movement direction. While this approach facilitates perception, the visibility may degrade in environments without such guidance, where only spontaneous eye movements are available. Third, this study focused on grayscale images, and extensions to color images and videos were not explored. Future work should investigate design methods suitable for each display format.

6 APPLICATION

The potential applications of the proposed method include the following scenarios. The first scenario is a directional information display that presents different visual content depending on the observer's movement direction. Since the proposed method allows different images to be perceived depending on the direction of eye movements, it becomes possible to switch the displayed information by projecting multiple images with different stripe orientations in adjacent regions. Such a display would be particularly useful in environments where the user's movement paths are partially predictable, such as exhibition spaces or public guidance systems.

The second scenario is an extended display that presents different images during eye fixation and during eye movements (Figure 8). In the basic configuration of the proposed method, only a gray uniform image is shown during fixation, and no meaningful image was presented. To address this limitation, we implemented an extended version of the method in which the pixel values of the complementary luminance components are adjusted so that the time-averaged luminance during fixation becomes another image. Although the perceived image during fixation was difficult to recognize with the naked eye, the switching of image content was clearly observed through a camera, demonstrating that different images can be presented depending on the observer's eye state.

7 CONCLUSION

In this study, we proposed a novel image decomposition method to enable grayscale representation in direction-dependent information displays using high-speed projection. By compressing the dynamic range of the target image and adding complementary luminance stripe patterns, we successfully extended the conventional binary display paradigm to support grayscale representation. Experimental results indicated that increasing N up to 8 significantly enhances visibility by enlarging the spatial interval between stripe components. Furthermore, we demonstrated an interactive application capable of presenting different meaningful images depending on whether the observer's gaze is stationary or in motion. Future work includes improving tone resolution, developing guidance-free visualization methods, and extending the approach to color and dynamic visual content.

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Number JP25K03164 and JST ASPIRE JPMJAP2404.

REFERENCES

- [1] P. G. J. Barten. *Contrast Sensitivity of the Human Eye and Its Effects on Image Quality*. SPIE Press, Bellingham, WA, 1999. 4
- [2] H. Benko and A. D. Wilson. Multi-point interactions with immersive omnidirectional visualizations in a dome. In *ACM International Conference on Interactive Tabletops and Surfaces, ITS '10*, p. 19–28. Association for Computing Machinery, New York, NY, USA, 2010. doi: 10.1145/1936652.1936657 1
- [3] H. de Lange Dzn. Research into the dynamic nature of the human fovea→cortex systems with intermittent and modulated light. i. attenuation characteristics with white and colored light. *J. Opt. Soc. Am.*, 48(11):777–784, Nov 1958. doi: 10.1364/JOSA.48.000777 2
- [4] H. Ikeda, T. Hayakawa, and M. Ishikawa. Bilateral motion display: Strategy to provide multiple visual perception using afterimage effects for specific motion. In *Proceedings of the 25th ACM Symposium on Virtual Reality Software and Technology, VRST '19*. Association for Computing Machinery, New York, NY, USA, 2019. doi: 10.1145/3359996.3364241 1
- [5] R. Miyazaki, S. Miyafuji, K. Kim, and H. Koike. A display method visible only during gaze fixation on moving objects using high-speed projection and striped pattern. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology, UIST Adjunct '25*. Association for Computing Machinery, New York, NY, USA, 2025. doi: 10.1145/3746058.3759000 1
- [6] R. Miyazaki, S. Miyafuji, and H. Koike. Be smart-phone zombie!: Guidance display for texting while walking using striped pattern and high-speed projection. In *Adjunct Proceedings of the 27th International Conference on Mobile Human-Computer Interaction, MobileHCI '25 Adjunct*. Association for Computing Machinery, New York, NY, USA, 2025. doi: 10.1145/3737821.3748531 1
- [7] F. Sakaue and J. Sato. Active 3d motion visualization based on spatiotemporal light-ray integration. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1977–1985, 2020. doi: 10.1109/CVPR42600.2020.00205 1
- [8] T. Sato, D.-H. Hwang, and H. Koike. Mliolight: Projector-camera based multi-layered image overlay system for multiple flashlights interaction. In *Proceedings of the 2018 ACM International Conference on Interactive Surfaces and Spaces, ISS '18*, p. 263–271. Association for Computing Machinery, New York, NY, USA, 2018. doi: 10.1145/3279778.3279797 1
- [9] M. Spindler, S. Stellmach, and R. Dachsel. Paperlens: advanced magic lens interaction above the tabletop. In *Proceedings of the ACM International Conference on Interactive Tabletops and Surfaces, ITS '09*, p. 69–76. Association for Computing Machinery, New York, NY, USA, 2009. doi: 10.1145/1731903.1731920 1